Time Centering

- **Year**: Centering at 2013=0 makes the intercept interpretable as the expected value at the study's start and avoids numerical issues that can arise with large year values (2013-2024). Also, including year^2 will allow us to study the (de)acceleration of the market over the decades.
    - Q: How does market value change annually compared to the 2013 baseline?

Grand Mean Centering

- **Age**: Market value is determined by biological career stage (e.g., "is Messi 29 and in his prime?" vs "is Messi 19 and a prospect?"). It does *not* matter if you are the oldest guy on a young team (relative age); the market still sees you as 29. We include the linear term (`age_c`) to catch the general trend and the squared term (`age_sq`) to capture the peak and decline curve.
    - Q: How does market value change as a player moves through their biological career arc (from prospect to prime to veteran)?

- **Foreign Player Proportion**: Since no club has 0% foreign players (lowest is 28%), we shift the zero point to the population average (Grand Mean) of internationalization. The intercept therefore represents a club with 'typical' diversity for the league.
    - Q: Does having a *higher proportion of foreign players than the average club* increase market value?

- **Height**: Height is a static physical trait. The market values a 195cm defender because he is objectively tall, not because he is taller than his teammates. Grand mean centering allows us to interpret the coefficient as "the premium for being 1cm taller than the average soccer player. Keeps the intercept interpretable as the value of an "average height" player.
    - Q: Does being taller than the average player in the league increase market value?

Year centering

- **UEFA** Coefficient = UEFA coefficients suffer from "grade inflation" or calculation changes over decades. A score of 15 in 2013 might mean something different than 15 in 2024. Centering by year isolates relative prestige, how dominant the club was that season compared to peers. It also removes the confounding effect of time-trends in the coefficient system itself.
  - Q: Does playing for a club that is currently outperforming the European average boost a player's value?

Year-Club mean centering (relative to teammates)

minutes_played_c: Market value is driven by a player's status within their team. This metric isolates whether a player is a "Starter" (positive value) or a "Bench Player" (negative value) relative to their specific squad in that specific season. It removes the noise of "teams that rotate more" vs "teams that don't," focusing purely on the player's hierarchy in the squad.

- Q: Does being a key player (playing more than the average teammate) increase market value?

Position centering:

- **Goals, Assists:** A goal scored by a striker is expected; a goal scored by a defender is rare and highly valuable. If we leave `goals` and `assists` as raw counts, the model treats 1 goal the same for everyone, potentially undervaluing defensive contributions. We will group by position as performance expectations vary entirely by role. When we center within the position group, we transform the variable from a raw count into a "performance above expectation" metric. This interprets as the effect of scoring above position expectations.
  - Q: Does scoring more goals than the average player in their position increase market value?" (This distinguishes a goal-scoring defender from an underperforming striker).

Non-centered fixed effects:

Performance Metrics

- **Red and Yellow card:** These variables have a meaningful 0 interpretation, and since our outcome is log transformed, these uncentered variables allow us to make percentage interpretations (i.e., one additional red card is associated with beta% change in market value).
    - Q: What is the direct market penalty for receiving one additional red/yellow card?

- **Position**: Reference group will be attackers.
    - Q: How much less is a defender valued compared to an attacker, holding performance constant?

- **Region**: Reference group European & Central Asia.
    - Q: What is the market premium for players in the 'Europe' region compared to South America?

- **Citizen**: Reference group is non-citizen of the country where the club is registered. The coefficient represents the simple difference in means between the two groups. A coefficient of +0.10 means "Citizens are valued 10% higher than Non-Citizens, all else held equal."
    - Q: Is there a homegrown premium' (or discount) for players who hold citizenship in the club's country compared to foreign players?

Random Intercept and slope

- Players: Yearly market values are nested within players so this makes them a clustering variable.
    - Q: After controlling for goals and age, how much variance in value is simply due to who the player is (marketing value/reputation)?

- Club: Players are nested within groups, but they can also be members of multiple clubs througout their career making this a secondary clustering variable.
    - Q: How much of the variation in player value is due to the club they play for ('Real Madrid Effect')?

- Year: Without this random slope, the model assumes a one-size-fits-all growth rate (parallel lines). By including it, we allow for the possibility that some clubs are incubators (where values skyrocket year-over-year) while others are stagnant (where values grow slowly or decline), even after controlling for player performance. The theoretical explanation is that elite clubs (e.g., Real Madrid, Manchester City) benefit from massive commercial exposure and "brand halos" that cause their players' values to inflate faster than the market average. Conversely, smaller clubs may see their players' values strictly tied to performance without the "inflationary" bonus of the club's brand over time. Without this random slope, we may underestimate the standard errors for time varying fixed effects.
    - Q: Does the rate of growth in player market value differ significantly between clubs?

Why cross-classified model?

Players can change teams over their span of their careers meaning that their player context can change. They may be a star player in one team and a bench player on a different team that they switch to. In a cross-classified model, the player is the constant unit (i), but the club (j) changes over time (t). If player (I) plays for club (j1) in year (t0) but switches to club (j2) in year (t1) the fixed effect in the model, such as minutes played, updates the players status, as opposed to having static or average predictors. Club attributes, such as foreign players and UEFA scores, ensures that when player move clubs, the model swaps the values of X_j1, for X_j2. Player's characteristics, for instance age and height, move with the player across time.

Assumptions about the data:

We log transform market value to 2024 USD because financial data follows a pareto distribution rather than a normal distribution. If we keep the outcome on the raw scale, our residuals will be extremely large for high value players, and small for low value players violating the assumption of heteroscedasticity. Log transforming the outcome compresses the extreme outliers by pulling the long right tail to try and make the distribution normal. We handle exchange rate fluctuation by using historical exchange rates to convert our values to USD.We also adjust market value to 2024 USD because 10 million in 2013 is not the same as in 2024 due to inflation, and by keeping the raw values we would lose real economic value. Therefore, this adjustment allows us to compare real economic power of a player in 2013 versus 2014 without the noise of currency fluctuation.

We assume that variables such as height and positions are time invariant for the duration of this study. At the club level, we assume mutually exclusiveness in that players can only belong to one club in a given year. Although a player could have switched clubs mid-season, we keep this constant for model simplicity. We assume that the relationship between age and log-market value is non-linear and parabolic so we introduce a quadratic term on age. Because we excluded observations with missing data (listwise deletion), we assume the data are Missing Completely at Random (MCAR) or that the proportion of missing cases (~2%) is negligible and does not introduce systematic bias.

Variable Type	Variable	Centering Strategy/ Form Factor	Justification
Time	Season Year	Min-Centering (+ Quadratic)	Establishes a meaningful starting intercept (2013 =0)
Contextual, Club	UEFA Coefficient	Year-Mean Centering	Isolates relative prestige; removes historical inflation
	Foreign Players %	Grand-Mean Centering	Moves baseline to "Average Club" (since 0% is unlikely)
Player Characteristics	Age	Grand-Mean (+ Quadratic)	Captures absolute career stage (Prospect $ \to$ Prime $\to$ Veteran)
	Height	Grand-Mean	Captures premium for absolute physical size
	Position	Reference group, attacker	Attackers tend to be valued more on average
Player Demographic	Citizen of Club Country	None (Raw binary 0/1)	Keeps "Non-Citizen" as clear reference group. Separates individual status from club composition (`foreign_players`)
	Region of origin	Reference group, Europe	Control variable for where player is from
Player Performance	Goals	Position-Centering	Preserves natural 0 and marginal value of 1 goal interpretation
	Assists	Position-Centering	Preserves natural 0 and marginal value of 1 assist interpretation
	Red Card	raw	Meaningful 0 interpretation
	Yellow Card	raw	Meaningful 0 interpretation
	Average minutes played in season	Year-Mean Centering, z-score	Isolates whether player is a starter or bench player